

National University of Computer and Emerging Sciences

Fast School of Computing Spring 2024

CS-1004 OOP (CS)

Part A

Quiz # 5&6 [10 marks]

Time = 20 minutes Name: _____

Roll No: _____

write output:

```
#include<iostream>
using namespace std;
class A{
public:
A(){ cout<<"A C"<<endl;}
~A(){ cout<<"A D"<<endl;}
};
class B{
A a1;
public:
B(){ cout<<"B C"<<endl;}
~B(){ cout<<"B D"<<endl;}
};
class C{
A * a;
B *b;
public:
C(A * a):a(a),b(new B){ cout<<"C C"<<endl;}
C(B* b):b(b),a(nullptr){cout<<"D C"<<endl;}
~C(){
    if(a!=nullptr)
        delete b;
    cout<<"C D"<<endl;
}
};
```

```
int main() {
A a;B b;
{
    C c(&a);
}
C c(&b);
cout<<"-----"<<endl;
return 0;
}
```

Output:

National University of Computer and Emerging Sciences

Fast School of Computing Spring 2024

CS-1004 OOP (CS)

Part B

Quiz # 5&6 [10 marks]

Time = 20 minutes Name: _____

Roll No: _____

write output:

```
#include<iostream>
using namespace std;
class A{
public:
A(){ cout<<"A C"<<endl;}
~A(){ cout<<"A D"<<endl;}
};
class B{
A a1;
public:
B(){ cout<<"B C"<<endl;}
~B(){ cout<<"B D"<<endl;}
};
class C{
A * a;
B *b;
public:
C(A * a):a(a),b(new B){ cout<<"C C"<<endl;}
C(B* b):b(b),a(nullptr){cout<<"D C"<<endl;}
~C(){
    if(a!=nullptr)
        delete b;
    cout<<"C D"<<endl;
}
};
```

```
int main() {
B b;A a;
{
    C c(&b);
}
C c(&a);
cout<<"-----"<<endl;
return 0;
}
```

Output: